

Experience	
Security Researcher , Trail of Bits, Brooklyn NY	Dec 2024 – August 2025
Summer Intern <i>C, Embedded, Static/Dynamic Analysis, Networking</i>	
- Discovered 21 vulnerabilities, including code execution, heap overflows, stack buffer overflows, null pointer de-references and wrote a 80 finds page report. - Audited OpenWRT - the largest open-source router firmware, focusing on custom C networking services and daemons (mdnsd, ubus, netifd, etc.) using manual code review, libFuzzer, AFL++, and fuzz-introspector; built complex IPC-emulating harnesses for targeted fuzzing, applied CodeQL and Semgrep, and adapted the Trail of Bits AIxCC Buttercups system to run locally with custom-written harnesses.	
Wintership <i>C++, Rust, Reverse Engineering, Windows, LDAP, Frida</i>	
Built an advanced Rust-based fuzzer using LibAFL with Frida mode, implementing grammar-based mutations defined via protobuf for the BER LDAP RFC and a custom mutator to bypass stateful checks; reverse engineered the Windows server-side LDAP API, performed dynamic analysis with API Monitor, x64dbg, and Frida, and developed C# executables and DLLs for dynamic hooking.	
Cloud Security Engineering , Datalink Networks – Santa Clara, CA	May 2023 – December 2024
Azure Entra ID, RBAC, Cloud Architecture, IAM Design, Zero Trust	
- Developed a new Zero Trust Model on Azure using NSGs, WAF, and ASGs for the internal and client Azure tenant in team of 3 to pass SOC-2 audit - Created a security dashboard using Power BI and Microsoft Defender Security APIs to highlight thousands of security issues that broke compliance standards. Performed Security Audits on Azure using results to conduct incident response to enhance system security of clients leading to upwards of 15% increase in secure score.	
Startup Software Engineer , Petition Power – Fullerton, CA	September 2023 – May 2024
Dart, Flutter, Python, REST API, Firebase, OAuth, Fullstack	
- Competed in Microsoft Imagine Cup as the youngest team, reaching the semi-finals and securing over \$10k in funding - Led a team of five to develop a mobile alternative to web-based petition platforms using a Build–Measure–Learn cycle; produced Gantt charts, data-flow, and user-flow diagrams to scope and deliver a production-ready MVP, authored a proposal and presentation for organizers, and designed and implemented both the frontend (Flutter) and backend (initially Firebase, later migrated to Azure) using MSSQL and the Django REST Framework.	
Activities	
CTF Player , Squid Proxy Lovers & SLICES [:] - Top 3 US 2025 and Top 23 Global	spl.team
Co Captain & Reverse Engineer - <i>IDA, Binary Ninja, Pwndbg, Angr, Z3, Jadx, AFL++, C, Rust, C++, Java, Python, Wireshark, x64dbg, IDApy</i>	
- Lead and manage team of over 30 members global to compete in international security completions consistently placing in the top 10 out of thousands of teams. Attended major finals, earned over \$40k in total team winnings, held large 20 person team onsite. Written 3 blog post regarding challenges and research - Personally focusing on Reverse Engineering, Misc & Hardware challenges dealing with self modifying code, Custom VMs, Custom CPUs, Custom/Esoteric Languages & Instruction sets, Mobile Applications, Compilers, Custom Address spaces schemes, library exploitation, zero-day hunting in modern applications. - Noteworthy Places: HITCON 2025 1st, Google CTF 3rd, Plaid CTF 2nd, Defcon Finals 3rd, Defcon Quals 10th, CSAW Finals 1st, 3rd ctftime.org/team/222966	
RIT Security Club	github.com/VRIG-Ritsec
Vulnerability Research Interest Group (VRIG) Co-Lead - <i>Swift, C++, Js, V8, Compilers, Vuln Research, Allocators, Fuzzing, Postgress, Docker</i>	
Led VRIG research advancing Google’s JavaScript fuzzer Fuzzilli by introducing novel feedback heuristics derived from V8 compiler internals, runtime-guided mutation strategies, and improved complex program generation, achieving 1.5–2× execution throughput; built a dockerized fuzzing network with batched synchronization to a global PostgreSQL backend for agentic analysis, and designed a multi-agent system for program template and corpus generation using runtime analytics to mitigate fuzzer plateaus and perform variant analysis, with extensive custom tool integrations for V8 and FAISS-based vector RAG.	
Reverse Engineering Lead	
- Taught reversing topics related to static analysis (IDA/Binary Ninja), dynamic analysis (Pwndbg, Frida, LD_PRELOAD, x64dbg, API monitor, Process monitor), APK/Android (Java, .NET, Dart), Constraint solving (Z3), Symbolic Execution Engines (Angr), and Virtual Machines. Creating and designing a rust ARM decompiler from scratch as a semester long class to identify and rebuild basic blocks, CFGs, loops, data flow, and deal with stack	
Squid Agent - spl.team/blog/squid-agent-csaw	Present
Developer and Team Lead - <i>Docker, Agentic Work Flows, IDApy, Web Exploitation, Cryptograph, Binary Exploitation, Reverse Engineering</i>	
- Achieved a solve rate of 92% (46/50) on CTFtiny benchmark and won first place at CSAW’s Agentic AI competition beating NYU’s Research system - DCIPHER (35/50). - Created several Multi-Agent CTF auto solvers optimized for each category with the a custom backend library and the ability to run <i>+100 challenges at once</i>. Implemented hundreds of custom tool calls to interact with security tools like IDA-Pro, Jadx, Pwndbg, Volatility. Allows system to fully autonomously solve all mainline ctf challenge categories and submit flags. - Engineered IkaCore, a hierarchical multi-agent LLM execution framework with graph-based staged workflows, provider-agnostic model abstraction, checkpoint/resume for long-running tasks, fine-grained tool invocation control, async execution, token-aware context management, rate limiting, and full lifecycle logging.	
Collegiate Penetration Testing Competition (CPTC) Member	Present
Finalist and Regional Winner 2nd & 3rd - <i>AzureHound, NMAP, enum4linux, Bloodhound, mimikatz, Impacket, HardenWindows, Prowler, CloudFox, Gobuster, metasploit</i>	
- Working in a team of 6 to perform enterprise pentesting specializing in Azure/AWS, AI/LLM , and Windows Active Directory based pentesting. Developed python and powershell scripts, tools and used open source tools to attack a fully simulated enterprise environment in a live situation, focusing on minimal interference and proper reporting of security and compliance based issues emulating a professional environment. Written 3 full reports (65-130 page) with 80 unique find total - Found several new zero days before/during the competition in open source software such as trafeik, librechat, keycloak. Allowing for SSRF, Arb File Write, dumping credentials via insecure APIs. Used a mix of manual code review, static analysis and agentic work flows to find, test and triage bugs.	
Other Awards	
CSAW Hardware Attack Finals, CyberBay \$25k, DiceCTF Finals, Hack the Code 2nd \$10k, lakeCTF Finalist, HTB Global Corporate 3rd, Cyberseed \$2.5k, Cypat 4x plat semis (13th)	